THE PRINCIPAL DIVISOR OF AN INVERTIBLE RATIONAL FUNCTION ON A PROPER INTEGRAL CURVE HAS DEGREE ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The graph of a rational function in a projective line bundle. Same chapter chow.tex, label lemma-rational-function, invoked at proof line 2826 and used throughout as 'all the results of this lemma'. Provides the diagram with q : Y -> P^1_K over Spec(K) and the identities div_X(f) = p_* div_Y(f) and div_Y(f) = [q^{-1}D_0]_0 - [q^{-1}D_infty]_0; the reduction to the pushforward along q is carried out in the counted span.
- Finiteness, flatness and finite locally free morphisms. Morphisms, labels morphisms-definition-finite-locally-free (proof line 2836) and morphisms-lemma-finite-flat (proof line 2862); same chapter chow.tex, label lemma-finite-in-codimension-one (proof line 2849). Supplies the criterion in dimension one that a dominant morphism from an integral curve to a regular curve is finite, and the equivalence between finite flat and finite locally free.
- Flatness of torsion free modules over a discrete valuation ring and the pushforward of a flat pullback. More on Algebra, label more-algebra-lemma-dedekind-torsion-free-flat, invoked at proof line 2858; Algebra, labels algebra-lemma-dim-affine-space (proof line 2854) and algebra-definition-ord (statement, proof line 2814); same chapter chow.tex, label lemma-finite-flat (proof line 2840). Shows every local ring of Y at a closed point is flat over the discrete valuation ring at its image because it is a domain, and computes the pushforward of the pulled back divisor as the degree times the difference of the two point classes.

Source reference. Original lemma 02MD: sources/stacks-chow/tags.txt, source lines 3440–3440. Navigation pointer for \ref{algebra-definition-ord} used inside the counted span (line 2814); the argument that consumes it is retained in the span.

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Source reference. Original lemma 00OP: sources/stacks-chow/tags.txt, source lines 903–903. Navigation pointer for \ref{algebra-lemma-dim-affine-space} used inside the counted span (line 2854); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RH: sources/stacks-chow/pinned-original-source.tex, source lines 2439–2439. Navigation pointer for \ref{lemma-finite-flat} used inside the counted span (line 2840); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RM: sources/stacks-chow/pinned-original-source.tex, source lines 2524–2524. Navigation pointer for \ref{lemma-finite-in-codimension-one} used inside the counted span (line 2849); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RQ: sources/stacks-chow/pinned-original-source.tex, source lines 2701–2701. Navigation pointer for \ref{lemma-rational-function} used inside the counted span (line 2826); the argument that consumes it is retained in the span.

Source reference. Original lemma 0AUW: sources/stacks-chow/tags.txt, source lines 13664–13664. Navigation pointer for \ref{more-algebra-lemma-dedekind-torsion-free-flat} used inside the counted span (line 2858); the argument that consumes it is retained in the span.

Source reference. Original lemma 02KA: sources/stacks-chow/tags.txt, source lines 3365–3365. Navigation pointer for \ref{morphisms-definition-finite-locally-free} used inside the counted span (line 2836); the argument that consumes it is retained in the span.

Source reference. Original lemma 02KB: sources/stacks-chow/tags.txt, source lines 3366–3366. Navigation pointer for \ref{morphisms-lemma-finite-flat} used inside the counted span (line 2862); the argument that consumes it is retained in the span.

Lemma 0.1. *Let K be any field. Let X be a 1-dimensional integral scheme endowed with a proper morphism $c : X \to \mathrm{Spec}(K)$. Let $f \in K(X)^*$ be an invertible rational function. Then*

$$\sum\nolimits_{x \in X \text{ closed}} [\kappa(x) : K] ord_{\mathcal{O}_{X,x}}(f) = 0$$

where ord is as in Algebra, Definition 02MD. In other words, $c_* div(f) = 0$.

Proof. Consider the diagram

$$\begin{array}{ccc} Y & \xrightarrow{p} & X \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle c} \\ \mathbf{P}^1_K & \xrightarrow{c'} & \mathrm{Spec}(K) \end{array}$$

that we constructed in Lemma 02RQ starting with X and the rational function f over $S = \mathrm{Spec}(K)$. We will use all the results of this lemma without further mention. We have to show that $c_*\mathrm{div}_X(f) = c_*p_*\mathrm{div}_Y(f) = 0$. This is the same as proving that $c'_*q_*\mathrm{div}_Y(f) = 0$. If $q(Y)$ is a closed point of $\mathbf{P}^1_K$ then we see that $\mathrm{div}_X(f) = 0$ and the lemma holds. Thus we may assume that q is dominant. Suppose we can show that $q : Y \to \mathbf{P}^1_K$ is finite locally free of degree d (see Morphisms, Definition 02KA). Since $\mathrm{div}_Y(f) = [q^{-1}D_0]_0 - [q^{-1}D_\infty]_0$ we see (by definition of flat pullback) that $\mathrm{div}_Y(f) = q^*([D_0]_0 - [D_\infty]_0)$. Then by Lemma 02RH we get $q_*\mathrm{div}_Y(f) = d([D_0]_0 - [D_\infty]_0)$. Since clearly $c'_*[D_0]_0 = c'_*[D_\infty]_0$ we win.

It remains to show that q is finite locally free. (It will automatically have some given degree as $\mathbf{P}^1_K$ is connected.) Since $\dim(\mathbf{P}^1_K) = 1$ we see that q is finite for example

by Lemma 02RM. All local rings of $\mathbf{P}^1_K$ at closed points are regular local rings of dimension 1 (in other words discrete valuation rings), since they are localizations of $K[T]$ (see Algebra, Lemma 00OP). Hence for $y \in Y$ closed the local ring $\mathcal{O}_{Y,y}$ will be flat over $\mathcal{O}_{\mathbf{P}^1_K,q(y)}$ as soon as it is torsion free (More on Algebra, Lemma 0AUW). This is obviously the case as $\mathcal{O}_{Y,y}$ is a domain and q is dominant. Thus q is flat. Hence q is finite locally free by Morphisms, Lemma 02KB. $\square$